



Produce a comprehensive research plan that uses causal discovery and causal inference to build on previous work with the SMMIS data set.

Based on your uploaded files — the full SMMIS variable list, the comprehensive literature review of Phil Steer's previous SMMIS work, and your causal inference and discovery references (Pearl's ladder of causation, Elements of Causal Inference, the Python causal inference textbook, and multiple algorithm survey papers) — here is a comprehensive research plan.

Comprehensive Research Plan: Causal Discovery and Causal Inference on the SMMIS Dataset

Context and Motivation

The SMMIS database covers 585,291 pregnancies from 15 hospitals in North West Thames (1988–2000), with prospectively collected data entered by trained midwives with online validation. Previous analyses by Steer and colleagues established important *associational* findings: a U-shaped relationship between maternal haemoglobin (Hb) and birth weight, an inverted-U relationship between diastolic blood pressure and birth weight, ethnic disparities in perinatal outcomes, and multi-factor intrapartum risk interactions. These studies relied primarily on regression and logistic modelling — rung-one (associational) reasoning on Pearl's Ladder of Causation. The proposed plan advances this work to rungs two (interventional) and three (counterfactual), using both causal *discovery* (learning the DAG structure from data) and causal *inference* (estimating effect sizes under that structure).^{[1] [2] [3]}

Phase 1: Problem Specification and Expert Knowledge Elicitation

Formulate Causal Questions

Rather than mining freely, begin with well-defined questions grounded in clinical theory. Candidate causal questions that directly build on previous SMMIS findings include:^[2]

- Does the lowest antenatal Hb (LOWHB) causally mediate the effect of maternal BMI (BMICD) on birth weight (BIRTHWGT)? Previous work showed both are independently associated with birth weight, but the causal pathway is unknown.^[3]
- Does the rise in diastolic BP (INCRBPCD) have a total causal effect on perinatal mortality (PERDTNCD) that is independent of birth weight? Previous work found a curvilinear

association but could not rule out confounding.^[3]

- Does ethnicity (ETHNIRCD) have a direct causal effect on adverse neonatal outcome (APGAR5, SCBU, PERDTNCD) after intervening on all upstream factors (BMI, Hb, gestational age, smoking)? Steer et al. found Black ethnicity remained significant after adjustment, but mediation structure is unresolved.^[3]
- Does meconium (MECONICD) directly cause adverse neonatal outcome, or is it a mediator/proxy for fetal hypoxia (confounded by gestation and ethnicity)?

Encode Background Knowledge

Use the temporal and biological ordering of SMMIS variables to construct a partial ordering that constrains discovery algorithms. This is a Tier-based background knowledge approach:^[4]

Tier	Variable Block	Rationale
1	ETHNIRCD, MARITACD, SPARENCD, CRSQUINT, MAGEDEL, BMICD	Fixed pre-pregnancy characteristics
2	SMOKINCD, HYPERTE, HISTDICD, PARITY, BOOKHB, MBOOKBP	Booking (first antenatal visit)
3	PREGDICD, UTICD, HIGHBP, PROTEICD, LOWHB, APHCD, PET, NOSCAN, NOANC, WGEST	Antenatal course
4	ONSETLCD, METHDECD, ANALAB, MECONICD, CTGCD, BLDLOSS, AUGMENCD	Intrapartum variables
5	BIRTHWGT, CENTILE, APGAR1, APGAR5, SCBUCD, PERDTNCD, NEODT28	Neonatal outcomes

No edge from a higher tier to a lower tier should be permitted. This background knowledge should be encoded in all discovery algorithms using prior knowledge modules (e.g., via `causal-learn`'s `BackgroundKnowledge` class).^[2]

Phase 2: Data Preparation

Variable Selection and Preprocessing

- Select the subset of variables marked γ (used in current dataset) from the SMMIS codebook, covering maternal demographics, booking variables, antenatal complications, intrapartum factors, and neonatal outcomes.^[5]
- Encode all categorical variables using the provided numeric recode scheme (SMOKINCD, METHDECD, OUTCOMCD, etc.).^[5]
- Handle missingness explicitly. SMMIS has non-ignorable missing data patterns (e.g., LOWHB missing when no low Hb was recorded). Use multiple imputation or treat missingness as a variable in its own right where clinically meaningful.
- Create derived variables as already done: BMICD, PET, INCRBPCD, DECRHBCD, PERDTNCD, WGEST.^[5]

- Restrict to singleton live births for primary analyses (as in previous work) to reduce heterogeneity.

Stratification Options

Consider running separate discovery analyses by:

- Nulliparous vs. multiparous (parity modifies almost all relationships)
- Pre-1995 vs. post-1995 (coding of some variables changed)
- Ethnic group subsets (to test whether causal structure differs across populations, a form of non-stationarity-based causal discovery)^[1]

Phase 3: Causal Discovery

The SMMIS data is observational i.i.d. cross-sectional data with a large sample size (~500,000 births), many mixed-type variables, likely non-linear relationships, and probable unmeasured confounders (e.g., socioeconomic factors beyond Carstairs index, maternal nutrition, paternal variables). These properties motivate the following algorithm selection strategy.

Algorithm 1: PC Algorithm (Causally Sufficient Baseline)

Apply the PC algorithm as a baseline, using Fisher's Z conditional independence test for continuous variables and appropriate chi-squared or G-tests for categorical/ordinal variables. The output is a CPDAG (Completed Partially Directed Acyclic Graph) representing the Markov Equivalence Class. Background knowledge tiers (Phase 1) are passed directly to constrain orientation.^{[6] [7] [1]}

- **Assumption check:** PC assumes causal sufficiency (no unmeasured confounders). This is unlikely to hold fully; use the result as a comparison baseline.^[7]
- **Implementation:** `causal-learn` library, PC class with `fisherz` or `kci` (Kernel-based CI test for non-linear relationships).^[2]

Algorithm 2: FCI Algorithm (Allowing Hidden Confounders)

Because unmeasured confounders almost certainly exist (e.g., genetic factors, nutrition, socioeconomic variables not captured in SMMIS), use the Fast Causal Inference (FCI) algorithm. FCI outputs a Partial Ancestral Graph (PAG) with bidirected edges indicating latent common causes.^{[8] [1]}

- FCI is asymptotically correct even in the presence of confounders.^[1]
- Use RFCI (fast version) for initial runs given the high dimensionality.^[1]
- The o-marks in PAG output will indicate which edges remain ambiguous (could be direct causation or hidden confounding) — clinically meaningful for generating hypotheses.
- Apply tiered background knowledge to FCI to improve orientation and reduce the equivalence class.^[4]

Algorithm 3: Score-Based Method — GES or GFCI

Use Greedy Equivalence Search (GES) as a complement to PC. GES starts from an empty graph and uses BIC scoring, making it less sensitive to individual conditional independence test errors — important given the large number of binary/categorical variables in SMMIS. GFCI (hybrid of GES and FCI) provides improved accuracy over FCI alone and is particularly recommended.^[1]

Algorithm 4: LiNGAM for Continuous Sub-problems

For the continuous variable sub-network (BMI, booking Hb, booking BP, highest BP, lowest Hb, birth weight, head circumference, birth length), apply **DirectLiNGAM**. LiNGAM exploits non-Gaussianity to identify the full DAG (not just a CPDAG), providing edge *direction* resolution within Markov equivalence classes. Given that biological variables such as Hb and BP are unlikely to be Gaussian, this is well-motivated.^{[7] [2] [1]}

Algorithm 5: DECI or NOTEARS (Continuous Optimisation, Large-Scale)

For the full variable set, run NOTEARS or DECI (gradient-based causal discovery) as a scalability solution. These methods formulate DAG learning as a continuous optimisation problem and scale well to dozens of variables. Use the discovered adjacency matrix as a starting skeleton for further constraint-based refinement.^{[7] [2]}

Sensitivity and Robustness

- Run all algorithms on bootstrap samples (e.g., 500 resamples) and report edge stability as the frequency with which each edge appears across bootstrap runs.^[8]
- Compare results across algorithms; edges that appear consistently across PC, FCI, and GES are most trustworthy.^[1]
- Use the structural Hamming distance (SHD) between algorithm outputs to quantify agreement.^[7]

Phase 4: Causal Inference (Effect Estimation)

Once a causal graph is established (or a set of candidate graphs), move to rung-two inference: estimating causal effects using the do-calculus.^{[4] [2]}

Step 1: Identifiability Check

For each target causal query (e.g., "what is the effect of LOWHB on BIRTHWGT?"), use the **backdoor criterion** or **front-door criterion** to determine whether the effect is identifiable from observational data. Encode the SMMIS DAG in DoWhy's `CausalModel` object and call `.identify_effect()` to obtain the estimand.^[2]

Step 2: Estimating Average Treatment Effects (ATE)

Backdoor Adjustment / Covariate Adjustment: For exposures where a valid adjustment set exists (identified from the DAG), use regression adjustment, stratification, or matching on the adjustment set.^[2]

Inverse Probability Weighting (IPW): For exposures with many confounders, use IPW to create pseudo-populations ^[2]. Estimate propensity scores ($P(\text{exposure} \mid \text{confounders})$) using logistic regression or gradient boosting, then weight outcomes. This is especially applicable to:

- Effect of smoking (SMOKINCD) on birth weight
- Effect of labour induction (ONSETLCD) on perinatal outcome
- Effect of analgesia type (ANALAB) on blood loss

Matching: Use coarsened exact matching (CEM) or approximate matching on the backdoor adjustment set for treatment comparisons (e.g., spontaneous vs. induced labour). Note that propensity score matching (PSM) should be avoided due to the PSM paradox when dimensionality is high.^[2]

Meta-Learners (S-Learner, T-Learner, X-Learner): Estimate Conditional Average Treatment Effects (CATEs) to identify heterogeneous effects — e.g., whether the effect of anaemia on birth weight differs by ethnic group or BMI category. Implement using DoWhy + EconML.^[2]

Step 3: Mediation Analysis

The SMMIS data contains a rich set of potential mediators. Key mediation questions:

- Does gestational age at delivery (WGEST) mediate the effect of low Hb on perinatal mortality? ^[3]
- Does birth weight mediate the effect of blood pressure trajectory on perinatal mortality?
- Does meconium mediate the effect of post-term gestation on adverse neonatal outcome?

Use the front-door criterion or structural mediation analysis with the identified DAG. Decompose total effects into direct and indirect effects using the `mediation` framework in DoWhy or the `causaleffect` R package.^{[4] [2]}

Step 4: Counterfactual Reasoning (Rung 3)

At the individual level, generate counterfactual predictions:^{[4] [2]}

- "What would this baby's birth weight have been if the mother's lowest Hb had been maintained above 105 g/L?"
- "Would this perinatal death have been avoided if labour had been induced one week earlier?"

These require fitting a Structural Causal Model (SCM) to the SMMIS data and using noise variable reconstruction. This is most tractable for linear sub-models but can be extended using additive noise models (ANMs).^{[4] [1]}

Step 5: Instrumental Variable Analysis

Several natural instruments exist in SMMIS or could be constructed:

- **Birth month / season** as an instrument for gestational age (established seasonal preterm birth pattern)^[3]
- **Hospital of delivery (HOSPITAL)** as an instrument for clinical practice variation (e.g., induction rates, epidural use)
- **Carstairs deprivation quintile (CRSQUINT)** as an instrument for social exposures

Apply two-stage least squares (2SLS) IV analysis where standard backdoor adjustment is insufficient due to unmeasured confounding, particularly for estimating the causal effect of labour management practices on neonatal outcomes.^[4]

Phase 5: Validation and Falsification

Refutation Tests

For every causal effect estimate, run DoWhy's refutation suite:^[2]

- **Add a random common cause** — checks that the estimate is not spuriously driven by unmeasured confounders
- **Placebo treatment test** — replace the true treatment with a random variable; the estimated effect should be ~zero
- **Data subset test** — check stability of estimates across random subsets of the data

Domain Validation

- Compare discovered graph edges against established obstetric knowledge. Edges contradicting known clinical pathways (e.g., a directed edge from birth weight *to* smoking) should be flagged as likely artefacts.
- Compare causal effect estimates against published randomised trial evidence where available (e.g., effect of epidural analgesia on labour outcomes).
- Validate ethnic disparity findings against natural experiment evidence from the literature.

Temporal Stability Check

Split the SMMIS data into two periods (1988–1994 and 1995–2000) and test whether the causal graph structure is invariant across periods using **Invariant Causal Prediction (ICP)**. Variables whose causal parents change across periods are likely subject to unmeasured policy confounding.^[4]

Phase 6: Software and Implementation Stack

Task	Recommended Tool
Causal discovery (PC, FCI, GES)	causal-learn(Python)
LiNGAM variants	lingam Python package
Continuous optimisation DAG	gCastle, NOTEARS
Causal inference (ATE, IPW, matching)	DoWhy + EconML
DAG visualisation and adjustment set identification	dagitty (R) or causal-learn visualiser
Mediation analysis	DoWhy mediation module
Bootstrap stability	Custom Python with networkx edge frequency tracking
Missing data	mice (R) or sklearn iterative imputer

Key Methodological Considerations

Given the SMMIS dataset's specific properties, several critical considerations apply throughout:

- **Mixed data types:** SMMIS contains continuous (BMI, Hb, BP, birth weight), ordinal (smoking level, perineal state), and binary variables. Use appropriate CI tests (Fisher's Z for Gaussian continuous, G-test or chi-squared for discrete, KCI for non-linear).^[1]
- **High dimensionality with large N:** The ~500K sample size is a major asset — it permits reliable high-order conditional independence testing and reduces false discovery rates in PC/FCI. However, the ~100-variable full set requires sparse graph assumptions (few parents per node).^[1]
- **Non-linearity:** The established U-shaped relationships (Hb–birth weight, BP–perinatal mortality) violate linear assumptions. Pearsonian correlation cannot capture these. Use KCI-based independence tests in PC/FCI, and non-linear ANMs or DECI for the full pipeline.^{[3] [2]}
- **Unmeasured confounders:** Genetic factors, nutrition, paternal variables, and detailed socioeconomic data beyond the Carstairs index are absent. FCI (not PC) is therefore the primary discovery algorithm, and IV methods complement adjustment-based inference.^[4]
- **Selection bias:** The SMMIS cohort covers one London region (1988–2000). Causal claims about mechanism are more generalisable than effect-size estimates, which may not transfer to other populations or time periods.^[8]

Expected Contributions

This programme of work would be the first systematic application of modern causal discovery algorithms to the SMMIS dataset. It would move beyond the associational findings of previous studies to produce:^[3]

1. A causal DAG of the maternal–fetal system as observed in a large urban UK cohort, providing a reusable scaffold for future analyses
2. Mediation-decomposed effect estimates for the Hb–birth weight, BP–perinatal mortality, and ethnicity–adverse outcome relationships
3. CATE estimates quantifying heterogeneous treatment effects of clinical interventions (induction, epidural, oxytocin) across subgroups
4. Counterfactual risk scores for individual pregnancies, directly enabling the AI-based intrapartum risk stratification tool proposed in the most recent SMMIS paper^[3]

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